

Swarm Daily NAV

Methodology & Dashboard

A decision-support marking framework for private-credit BDC positions, combining benchmark spread movements with a daily news-event idiosyncratic overlay. Pinned per row in `daily_marks.methodology_version` so every mark is reproducible.

Prepared for executive review — 2026-05-17

Not a 40-Act fair-value mark. Decision-support only.

1. What this is

Swarm produces a **daily decision-support mark** for every borrower in the GSCR and GSBD BDC portfolios (~617 borrowers, ~2,300 individual positions). Unlike the quarterly fair-value mark required by the 40-Act, this daily mark is built for in-quarter risk management: it triangulates today's price move from three independent signals and flags borrowers that drift materially from their last reported FV.

The output sits behind the **Daily NAV** tab on the dashboard. Each row is clickable and opens a methodology drawer that shows every input that produced the mark — the benchmark spread moves, the DCF/market blend, any idio overlay, and which rails (if any) fired.

Three pillars

Pillar A — DCF	Discounts the position's cash flows using the high-yield OAS curve as a proxy for the obligor's discount rate. Picks up changes in the broad credit-cycle discount rate.
Pillar B — Market comps	Weighted basket of public benchmark moves (HY OAS series, sector ETFs, BDC ETFs) calibrated to the borrower's industry, seniority, and duration. Picks up sector- and rating-specific price moves.
Idiosyncratic overlay	News-driven shock when an event (bankruptcy, default, downgrade, covenant breach, etc.) fires on the borrower. Captures borrower-specific risk that benchmarks miss.

2. How a daily mark is calculated

Every weekday at 15:00 UTC, the mark-positions cron reads the prior day's fair value for each position, pulls today's benchmark snapshots, applies the methodology, and writes a new row to `daily_marks`. Each row carries a `JSONB components` field with every input value, so the mark is fully reproducible.

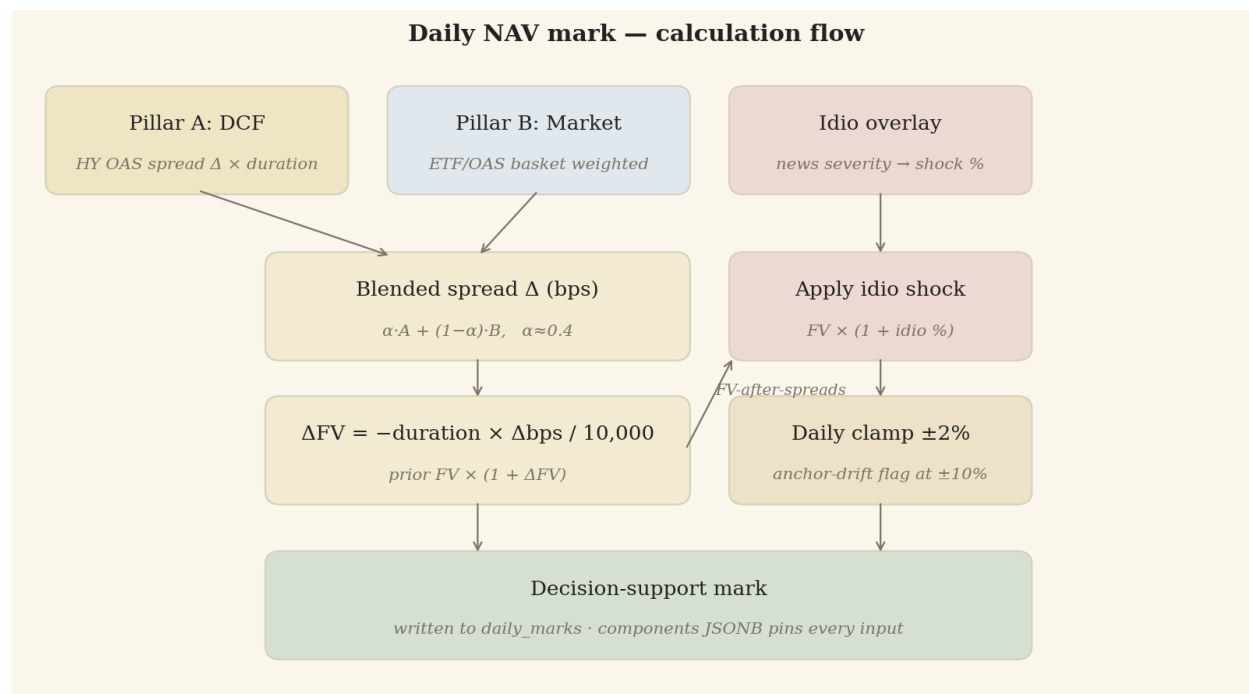


Figure 1. Daily mark calculation flow.

Step-by-step

Pillar A — DCF spread Δ	Today's change in the BAML HY OAS index, in bps. Direct read from the FRED snapshot.
Pillar B — market-comp spread Δ	For each benchmark in the position's weight map, convert today's price change to an implied spread Δ (using a duration proxy for ETFs, raw delta for yield series), then weight by the position's mapped weight. Sum to get pillar B in bps.
Blend	$\text{blended } \Delta \text{ bps} = \alpha \cdot \text{pillar_A} + (1 - \alpha) \cdot \text{pillar_B}$. α (the DCF weight) is ~0.4 by default, tunable per industry via the tuner.
FV move from spreads	$\text{FV move \%} = -\text{duration} \times \text{blended_bps} / 10,000$. Apply: $\text{FV} = \text{prior_FV} \times (1 + \text{FV_move_}\%)$.

Idiosyncratic overlay	If a news event fired on this borrower in the last 5 days with severity ≥ 70 , apply $FV \leftarrow FV \times (1 + \text{idio_shock_}\%)$. Severity-to-shock mapping is shown in Figure 3.
Daily clamp	Hard cap at $\pm 2\%$ intraday move vs. prior FV. Anything beyond clamps and sets a rail flag.
Anchor-drift check	Compare today's modeled FV against the last reported FV (the 'anchor', from the most recent quarterly observation). Drift $> 10\%$ sets a review flag — borrower needs PM attention.

3. News-event idiosyncratic signal

The benchmark pillars capture *market* moves but miss *borrower-specific* events — a covenant amendment, a 8-K bankruptcy filing, a downgrade announcement. The news pipeline ingests headlines from three sources every morning before the marking cron runs, scores them deterministically, and writes severity ≥ 70 events into the `detector_hits` table that the marking runner already consumes.

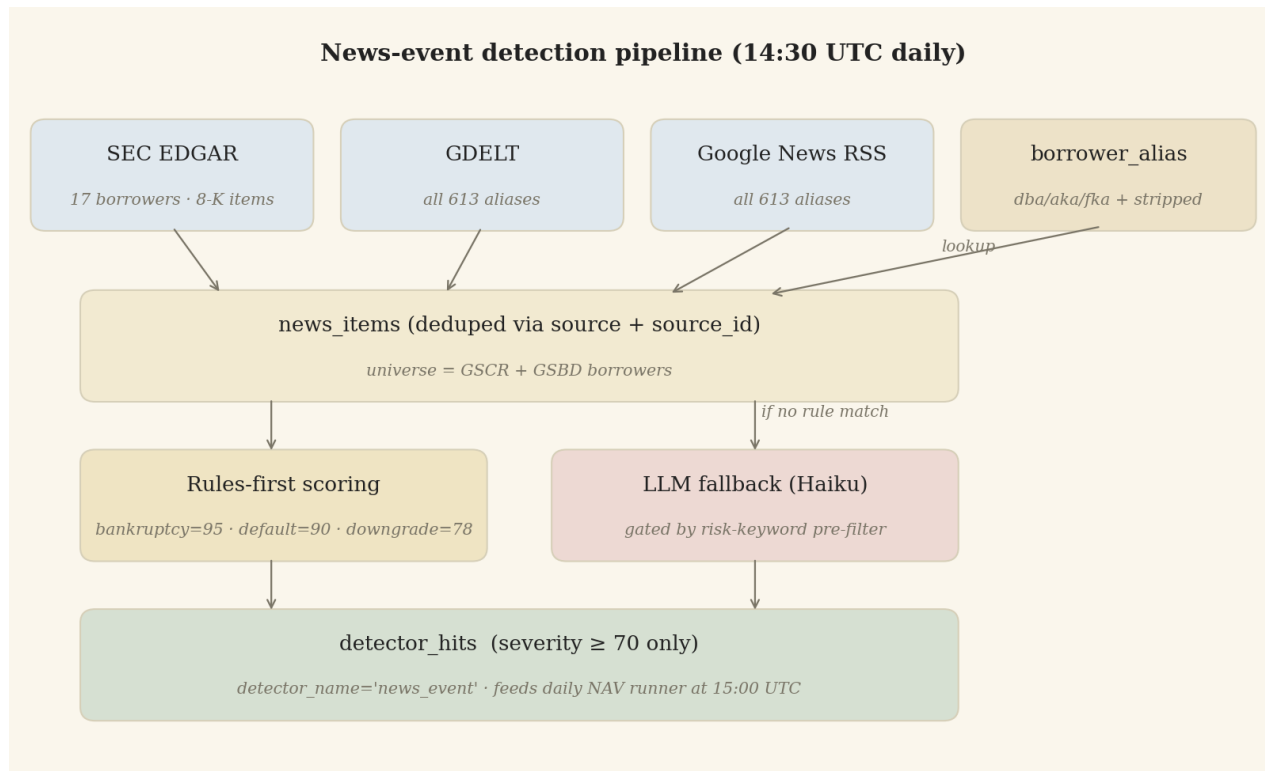


Figure 2. News-event detection pipeline. Runs at 14:30 UTC, 30 min before the marking cron.

Sources

SEC EDGAR 8-K filings — high-precision structured signal for the ~17 borrowers in the universe that are public filers or maintain SEC registration for high-yield bondholders. Each 8-K's filed item codes map deterministically to severity (Item 1.03 bankruptcy → 95, Item 4.02 restatement → 82, etc.).

GDELT — free generalist news indexer. Queries each borrower alias once per day. Coverage of private borrowers is patchy; included for breadth.

Google News RSS — free, broader trade-press coverage. Catches the private-LBO news that GDELT frequently misses (lender-letter coverage, covenant-amendment leaks, distress reporting).

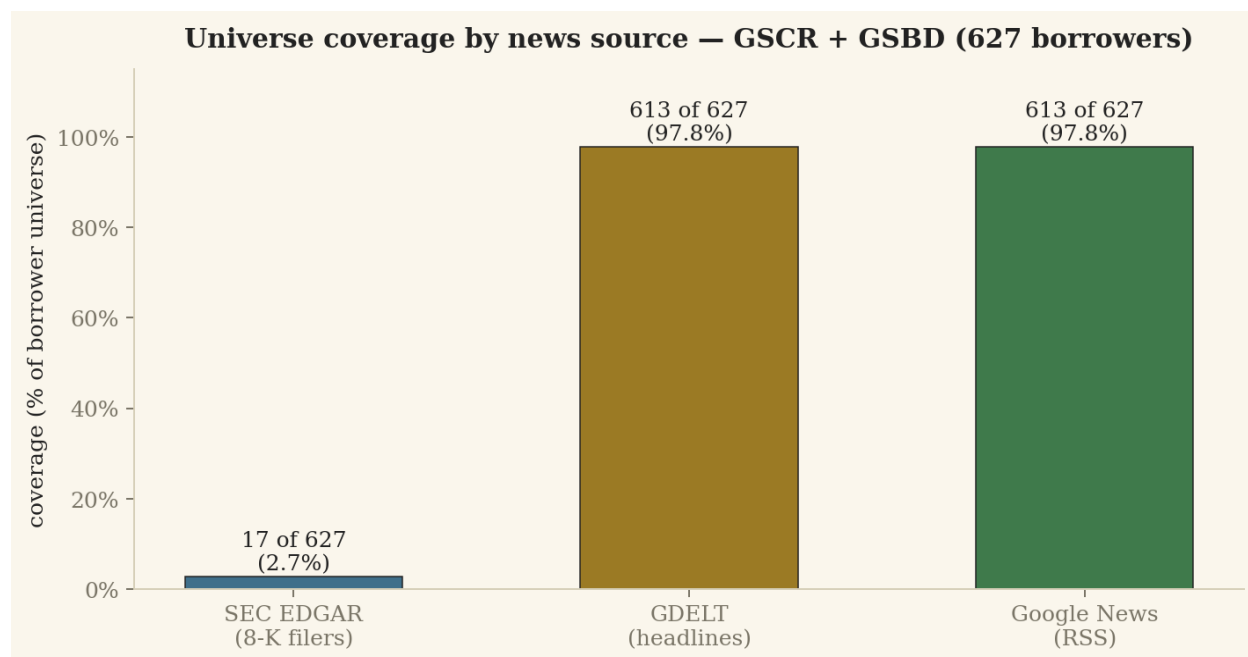


Figure 3. Coverage of the borrower universe by news source.

4. From severity score to idio shock

Each news event is scored on a 0–100 severity scale, calibrated against the kinds of credit events PMs care about. Scoring happens in two passes:

Rules-first — deterministic regex over the headline ('chapter 11', 'payment default', 'covenant breach', 'downgrade', 'lay off', etc.) and 8-K item codes. Free, predictable, ~70% of events resolve here.

LLM fallback — anything not matched by rules but containing a risk-adjacent keyword AND mentioning the borrower name is sent to Claude Haiku for a {severity, reason} response. Bounded by a keyword pre-filter to keep costs sub-\$15/month.

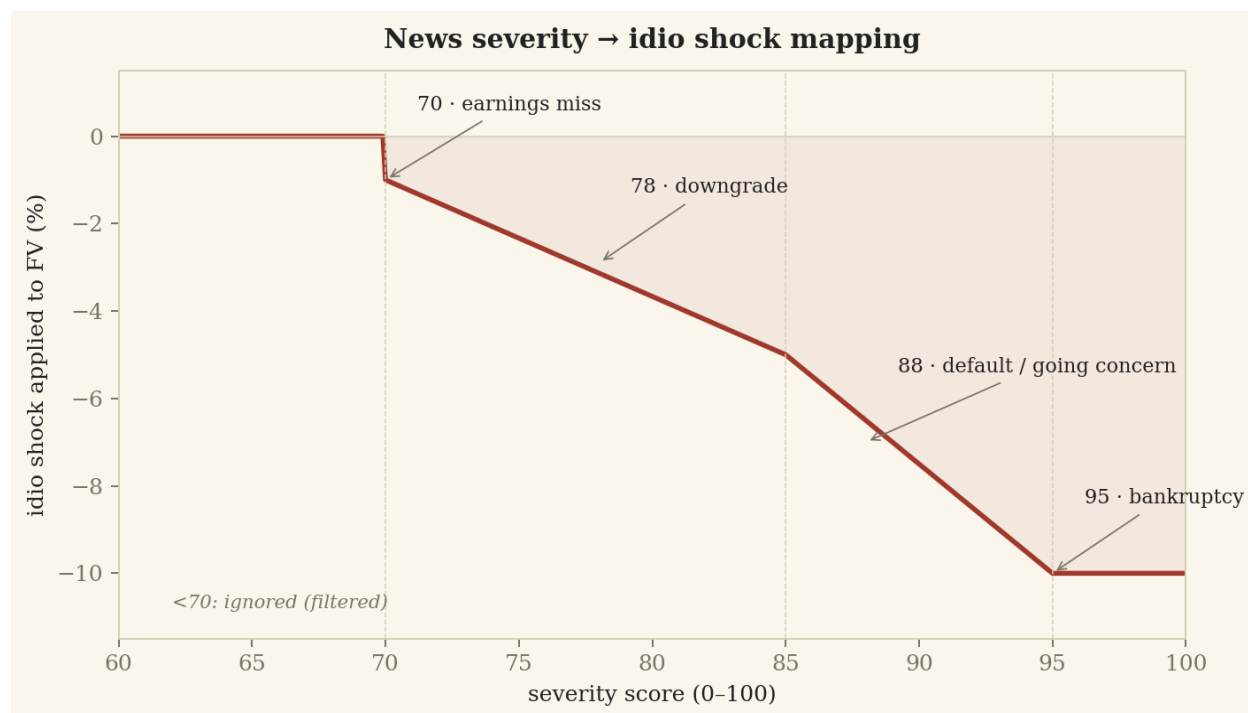


Figure 4. Severity score → idio shock applied to fair value. Events below severity 70 are ignored entirely.

Why these anchor points

Bankruptcy and default are unambiguous credit events — a −5% to −10% intraday mark is consistent with how dealers actually trade distressed paper. Downgrades and layoffs are softer signals that warrant a smaller adjustment. Anything below 70 is ambiguous enough that we'd rather miss a real signal than create false alarms.

5. What the dashboard shows

Daily NAV tab

The main table lists every position with today's mark, the Δ bps vs. prior, a 30-day mark trajectory sparkline, and a confidence flag (low/med/high). Sortable by borrower, fund (GSCR/GSBD), today's FV, Δ bps, mark %, confidence, or review flag. Filters: down today / up today / review queue.

Visual cues: red Δ bps for moves < -150 bps; amber for -150 to -50 ; the pillar bar visualises how today's move decomposes across DCF, market-comp, and any idio shock. A row flagged for review (rail fired, anchor drift $> 10\%$, or new idio shock) routes to the review queue.

Methodology drawer

Clicking any row opens a drawer showing the complete component breakdown:

- Prior FV, today FV, anchor FV, anchor drift % — the four key numbers, with thousand separators.
- Spread delta triangulation — α weight, position duration, pillar A bps, pillar B bps, blended bps.
- Benchmark inputs — every series in the weight map with its prior, today, and Δ bps.
- Overlay + rails — idio shock %, daily floor/ceiling fired (yes/no), anchor-drift flag.
- Overrides — list of any PM-submitted manual overrides (audit-only — never mutates the source mark), plus an override form.

Override workflow

When the model is wrong (it will be — illiquid private credit isn't a solved problem), the PM can submit an override from the drawer: override %, reason (free text), approver name. Overrides are **audit-only**: they're written to a separate `mark_overrides` table and never modify `daily_marks`. Each override moves through pending $\rightarrow$ approved/rejected. The full audit trail is preserved.

6. Governance & limitations

What this is not

This is **not a 40-Act fair-value mark**. The 40-Act mark continues to come from the quarterly board-approved valuation process. The daily mark is decision-support — it surfaces moves that warrant attention between quarterly marks and gives PMs a defensible in-quarter view.

Rails & flags

Three governance rails are baked into every mark:

Daily clamp $\pm 2\%$ — caps intraday moves. A model that wants to move a position 5% in a day gets clamped to 2% and the row is flagged for review. Prevents data-error blow-ups from contaminating the book.

Anchor drift $> 10\%$ — when the cumulative modeled drift from the last reported FV exceeds 10%, the row is flagged. This catches cases where the anchor is stale (pre-downgrade) and the model is correctly diverging.

Idio overlay fires $\rightarrow$ low confidence — any row with a non-zero idio shock is automatically downgraded to 'low' confidence so PMs know not to trust it blindly.

Reproducibility

Every `daily_marks` row pins the methodology version it was produced with. Old marks remain reproducible because the formula version is captured alongside the inputs. When we tune α or change benchmark weights, the new version writes alongside the old; no historical marks are modified.

Cost

All free except the optional LLM fallback for news classification, which is gated by a keyword pre-filter and runs against Claude Haiku 4.5. Realistic monthly cost $\leq \$15$ even with the full universe. Without the LLM branch (rules-only) the system runs at zero variable cost.

Coverage caveats

Of ~ 617 borrowers, only ~ 17 are SEC filers, so the high-precision EDGAR branch covers $\sim 3\%$ of the universe. The remaining $\sim 97\%$ rely on GDELT and Google News, which are noisier but free. The natural next step if PMs want higher signal-to-noise is a paid trade-press feed (9fin, Reorg) for the distressed-watchlist subset of borrowers.